Creating Training and Validation Data

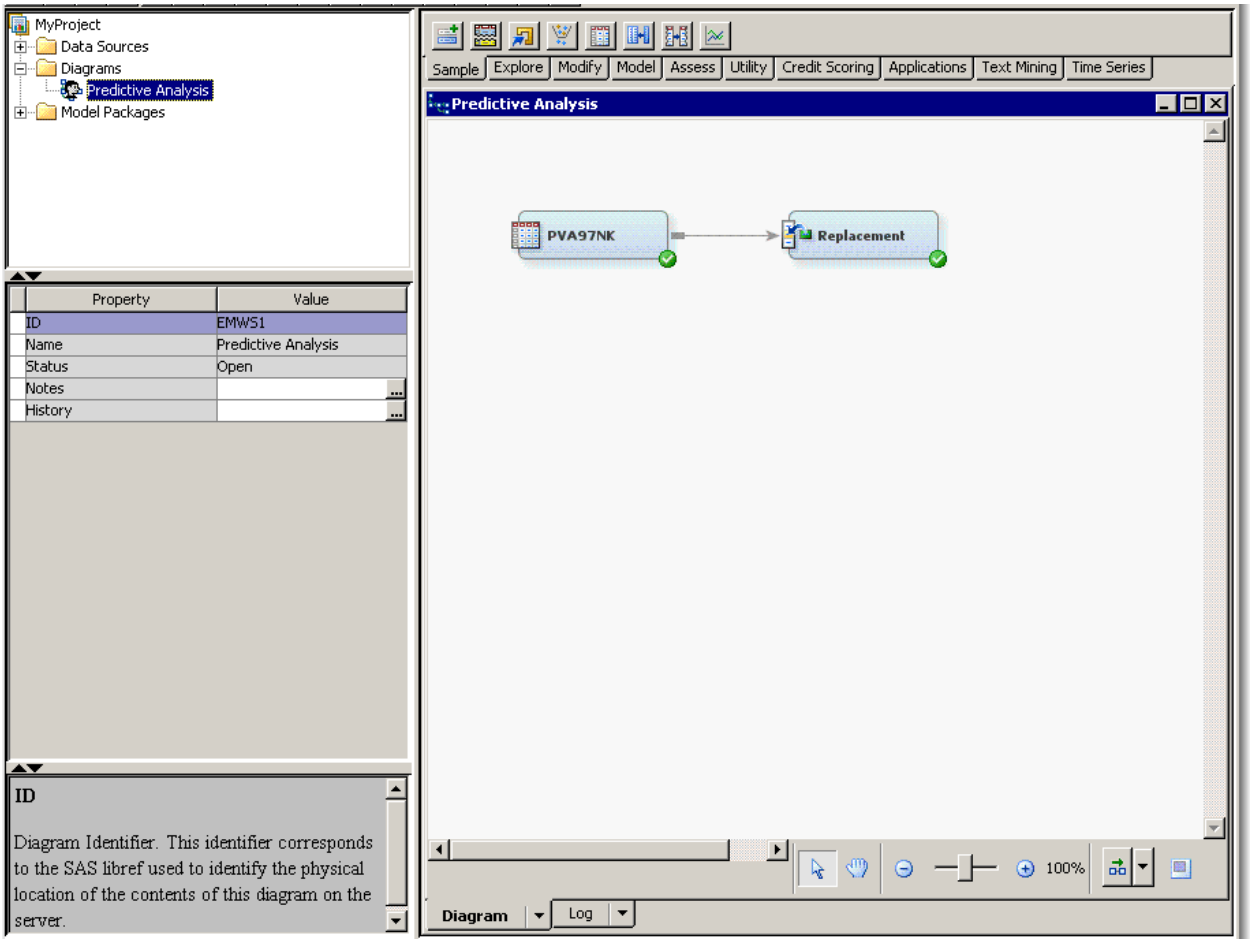
A critical step in prediction is choosing among competing models. Given a set of training data, you can easily generate models that very accurately predict a target value from a set of input values.

Unfortunately, these predictions might be accurate only for the training data itself. Attempts to generalize the predictions from the training data to an independent, but similarly distributed, sample can result in substantial accuracy reductions.

To avoid this pitfall, SAS Enterprise Miner is designed to use a *validation data set* as a means of independently gauging model performance. Typically, the validation data set is created by partitioning the raw analysis data. Cases selected for training are used to build the model, and cases selected for validation are used to tune and compare models.

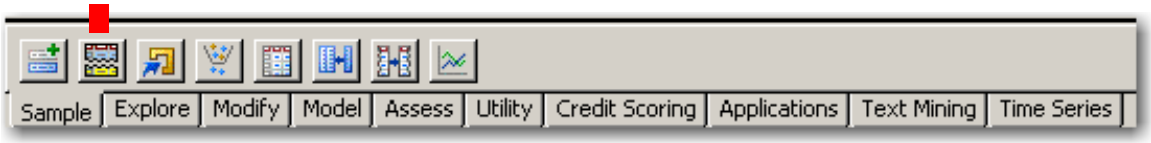
SAS Enterprise Miner also enables the creation of a third partition, named the *test data set*. The test data set is meant for obtaining unbiased estimates of model performance from a single selected model.

Your SAS Enterprise Miner workspace should appear as shown below:



Follow these steps to partition the raw data into training and validation sets:

1. Select the **Sample** tool tab. The Data Partition tool is the second from the left.



2. Drag a **Data Partition** tool into the diagram workspace, next to the Replacement node.

3. Connect the **Replacement** node to the **Data Partition** node.

4. Select the **Data Partition** node and examine its Properties panel.

Property	Value
General	
Node ID	Part
Imported Data	
Exported Data	
Notes	
Train	
Variables	
Output Type	Data
Partitioning Method	Default
Random Seed	12345
Data Set Allocations	
Training	40.0
Validation	30.0
Test	30.0
Report	
Interval Targets	Yes
Class Targets	Yes

Use the Properties panel to select the fraction of data devoted to the Training, Validation, and Test partitions. By default, less than half the available data is used for generating the predictive models.

There is a trade-off in various partitioning strategies. More data devoted to training results in more stable predictive models, but less stable model assessments (and vice versa). Also, the Test partition is used only for calculating fit statistics after the modeling and model selection is complete. Many analysts regard this as wasteful of data.

A typical partitioning compromise foregoes a Test partition and places an equal number of cases in Training and Validation.

5. Type **50** as the Training value in the Data Partition node.

6. Type **50** as the Validation value in the Data Partition node.

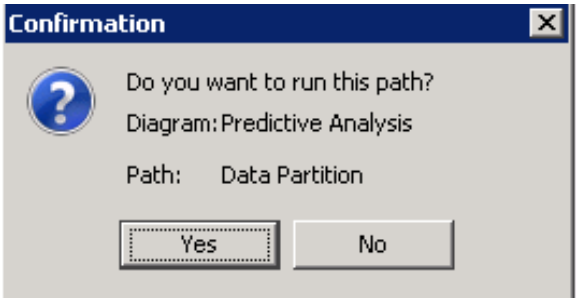
7. Type **0** as the Test value in the Data Partition node.

Data Set Allocations	
Training	50.0
Validation	50.0
Test	0.0

With smaller raw data sets, model stability can become an important issue. In this case, increasing the number of cases devoted to the Training partition is a reasonable course of action.

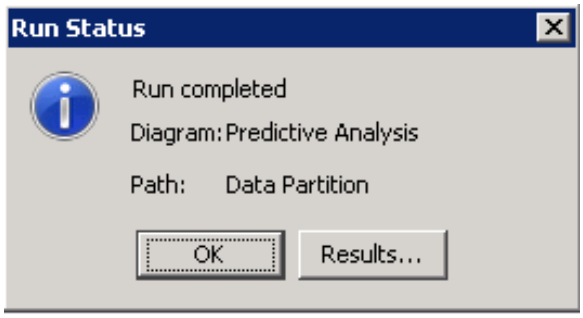
8. Right-click the **Data Partition** node and select **Run**.

9. Select **Yes** in the Confirmation window. SAS Enterprise Miner runs the Data Partition node.



Only the Data Partition node should run, because it is the only "new" element in the process flow. In general, SAS Enterprise Miner only runs elements of diagrams that are new or that are affected by changes earlier in the process flow.

When the Data Partition process is complete, a Run Status window appears.



10. Select **Results** in the Run Status window to view the results.

The Results - Node: Data Partition window provides a basic metadata summary of the raw data that feeds the node and, more importantly, a frequency table that shows the distribution of the target variable, **TARGET_B**, in the raw, training, and validation data sets.

Summary Statistics for Class Targets

Data=DATA

Variable	Numeric Value	Formatted Value	Frequency Count	Percent	Label
TARGET_B	0	0	4843	50	Target Gift Flag
TARGET_B	1	1	4843	50	Target Gift Flag

Data=TRAIN

Variable	Numeric Value	Formatted Value	Frequency Count	Percent	Label
TARGET_B	0	0	2422	50.0103	Target Gift Flag
TARGET_B	1	1	2421	49.9897	Target Gift Flag

Data=VALIDATE

Variable	Numeric Value	Formatted Value	Frequency Count	Percent	Label
TARGET_B	0	0	2421	49.9897	Target Gift Flag
TARGET_B	1	1	2422	50.0103	Target Gift Flag

The Partition node attempts to maintain the proportion of zeros and ones in the Training and Validation partitions. The proportions are not exactly the same due to an odd number of zeros' and ones' cases in the raw data.